$$= [x^4]' (e^x) + (x^4) [e^x]'$$

$$= 4x^3e^x + x^4e^x$$

$$= (4x^3 + x^4)e^x$$

(iv) Let $y = \frac{x^2-1}{x+3}$

This is of the type

$$\left[\frac{u}{v}\right]' = \frac{u'v - uv'}{v^2}$$

$$\frac{dy}{dx} = \left[\frac{x^2-1}{x+3}\right]'$$

$$= \frac{(x^2-1)'(x+3) - (x^2-1)(x+3)'}{(x+3)^2}$$

$$= \frac{(2x)(x+3) - (x^2-1)\times 1}{(x+3)^2}$$

$$= \frac{2x^2+6x-x^2+1}{(x+3)^2}$$

$$= \frac{x^2+6x+1}{(x+3)^2}$$

## Repeated differentiation:

If the derivative of a function is again differentiated with respect to the same variable, we say that the differentiation is the second order differentiation and is denoted as $\frac{d}{dx}(\frac{dy}{dx}) = \frac{d^2y}{dx^2}$ *or* $D^2y$ *or* $Y_2$

For example if $y = (x^3+4x^2+7)$ , then $\frac{dy}{dx} = (3x^2+8x)$

Again differentiating with respect to $x$, we get

$$\frac{d^2y}{dx} = \frac{d}{dx}(\frac{dy}{dx}) = \frac{d}{dx}(3x^2 + 8x)$$

$$= 6x+8$$

**NOTE**

Repeated or successive differentiation may be extended to any higher order derivation.

### 7.5.2 Integration

Integration is the reverse process of differentiation. It is also called anti-derivative.

Suppose the derivative of $x^5$ is $5x^4$. Then the integration of $5x^4$ with respect to x is $x^5$. we use this in symbol as follows: